Counterfactual Explanations for Multivariate Time Series Forecasting : A Unified Framework and Empirical Evaluation

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***Abstract*—Deep learning models achieve state-of-the-art performance in multivariate time series forecasting, yet their black-box nature limits their adoption in high-stakes applications requiring transparency and trust. Counterfactual explanations provide instance-level interpretability by identifying minimal changes to past observations that lead to a desired prediction outcome.**

We propose a unified formal framework for counterfactual explanations in multivariate forecasting, defining consistent notions of validity, minimality, and temporal plausibility for continuous-valued predictions. Within this framework, we conduct a systematic empirical evaluation of ForecastCF, a gradient-based optimization method, across four benchmark datasets and three neural forecasting architectures : DLinear, GRU, and N-BEATS.

Our results show that predictive accuracy does not correlate with counterfactual quality, revealing a structural gap between forecasting performance and explainability. We further analyze the impact of target tolerance intervals on counterfactual feasibility and stability. These findings provide practical insights and establish a foundation for future research on actionable counterfactual explanations in time series forecasting.

***Index Terms*—Explainable artificial intelligence, counterfactual explanations, time series forecasting, multivariate time series, interpretable machine learning, optimization-based methods**

I. Introduction

Accurate forecasting in multivariate time series is critical for high-stakes domains such as healthcare, finance, and energy management. Deep learning models have become the dominant approach due to their ability to capture complex temporal dependencies and cross-variable interactions. However, their black-box nature limits their adoption in practice, as stakeholders require not only accurate predictions but also interpretable and actionable insights.

Explainable Artificial Intelligence (XAI) methods for time series have mainly focused on global feature importance [1] or local saliency maps [2]. While informative, these approaches fail to address the fundamental "what-if" questions required for decision-making. Counterfactual (CF) explanations overcome this limitation by identifying minimal changes to input features that lead to a desired prediction outcome [3]. In forecasting settings, this corresponds to generating alternative historical trajectories that produce different future predictions (see Fig. 1).

Despite their success in classification, counterfactual explanations for time series forecasting introduce specific challenges. First, the output space is continuous, making target specification non-trivial. Second, temporal dependencies induce complex propagation effects, where local perturbations influence the entire forecast horizon. Third, existing work lacks a unified formal framework to consistently define and evaluate counterfactual explanations across models and datasets.

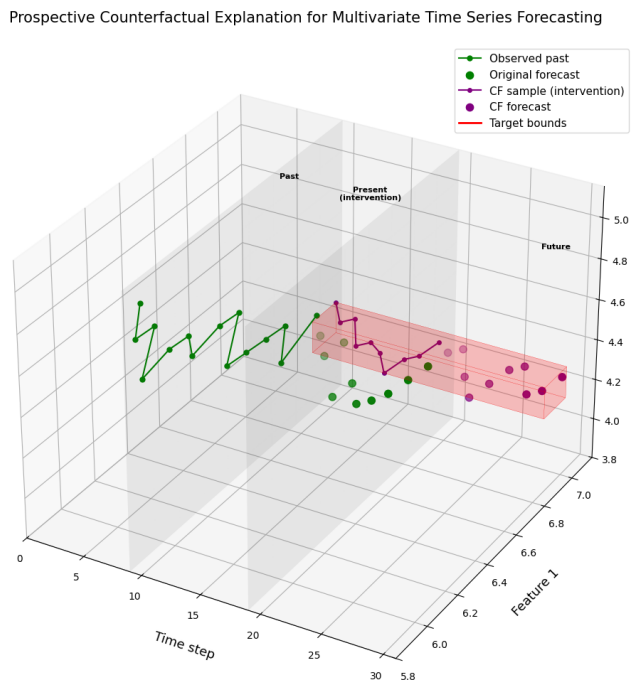

Figure 1. Counterfactual explanation in a forecasting setting. Minimal modifications to historical observations lead to an alternative forecast trajectory satisfying predefined target bounds.

In this work, we propose a unified formal framework for counterfactual explanations in multivariate time series forecasting and provide an empirical evaluation on univariate forecasting settings as a controlled benchmark. Our contributions are threefold :

- We introduce a formal framework defining validity, proximity, and compactness for counterfactual explanations with

continuous-valued targets.

- We analyze and compare counterfactual generation strategies, including gradient-based and instance-based approaches, and introduce simple baseline methods.
- We conduct a systematic empirical evaluation of ForecastCF [4] across multiple datasets and architectures (DLinear, GRU, N-BEATS), revealing an accuracy-explainability gap and analyzing the impact of target tolerance.

The remainder of this paper is organized as follows. Section II presents the formal framework. Section III reviews counterfactual generation methods. Section IV describes the experimental setup, Section V discusses the results, and Section VI concludes the paper.

II. Formal Framework

This section introduces a unified formal framework for counterfactual reasoning in time series forecasting. The notation is defined in the multivariate setting to provide a general and robust theoretical foundation, independently of the dimensionality used in the empirical evaluation.

A. Notation and Time Series Definition

Throughout this paper, vectors are denoted by underlined lowercase letters ($\underline{x}$), scalars by lowercase letters (x), and matrices by uppercase letters (X).

A *multivariate time series* is defined as an ordered sequence of T observations $X = (\underline{x}_1, \dots, \underline{x}_T) \in \mathbb{R}^{T \times d}$, where each vector $\underline{x}_t = (x_t^{(1)}, \dots, x_t^{(d)})^\top$ represents d co-variates observed at time t.

B. Forecasting Models

Given an observed history $X \in \mathbb{R}^{T \times d}$ and a forecast horizon $H \in \mathbb{N}^*$, we distinguish between two modeling paradigms :

- **Deterministic Forecasting :** A model f defines a mapping $f : \mathbb{R}^{T \times d} \to \mathbb{R}^{H \times d}$ such that $\widehat{Y} = f(X)$, where $\widehat{Y} = (\widehat{\underline{x}}_{T+1}, \dots, \widehat{\underline{x}}_{T+H})$ represents the future trajectory predicted.
- **Probabilistic Forecasting :** A model f estimates the conditional distribution $P(\underline{X}_{T+1:T+H} \mid \underline{X}_{1:T} = X)$, explicitly capturing predictive uncertainty [5].

In the following, we focus on deterministic neural forecasting models (DLinear, GRU, N-BEATS), as gradient-based counterfactual methods require access to model derivatives.

C. Counterfactual Targets and Validity

In contrast to classification tasks with discrete outputs, forecasting produces continuous-valued predictions, making the definition of a counterfactual target non-trivial. We formalize the desired outcome through a Boolean condition.

Let $C : (\mathbb{R}^d)^H \to \{0, 1\}$ be a Boolean condition defined on the forecast outcomes. The **counterfactual target set** is defined as

$$\mathcal{Y}_{\text{target}} = \{\, Y \in (\mathbb{R}^d)^H : C(Y) = 1 \,\}.$$

A counterfactual explanation X' is said to be **valid** if it modifies the model output so that it belongs to the target set. Formally, validity is defined as :

1) $f(X') \in \mathcal{Y}_{\text{target}}$ for deterministic models;
2) $P_f(\mathcal{Y}_{\text{target}} \mid \underline{X}_{1:T} = X') \geq \tau$ for probabilistic models, where $\tau \in [0, 1]$ is a confidence threshold.

A *counterfactual explainer* is defined as a mapping

$$E : (\underline{x}_{1:T}, f, \mathcal{Y}_{\text{target}}) \mapsto \underline{x}'_{1:T}$$

that returns a valid counterfactual.

D. Desirable Properties

Beyond validity, a counterfactual explanation X' should satisfy several properties to ensure both interpretability and practical relevance :

- **Proximity :** The counterfactual should remain as close as possible to the original input X, ensuring minimal deviation. This is typically measured using the L_2 norm :

$$dist(X, X') = \sum_{t=1}^{T} ||\underline{x}'_t - \underline{x}_t||_2.$$

- **Compactness :** The number of modified timesteps should be limited to reduce cognitive complexity. In practice, a trade-off often exists between achieving validity and maintaining high compactness.
- **Temporal Plausibility :** The counterfactual trajectory should preserve realistic temporal dynamics. This can be encouraged by penalizing deviations in temporal increments :

$$||\Delta \underline{x}'_t - \Delta \underline{x}_t||_2,$$

where $\Delta \underline{x}_t = \underline{x}_{t+1} - \underline{x}_t$.
- **Actionability :** In multivariate settings, modifications should be restricted to a subset of actionable co-variates $\mathcal{A} \subseteq \{1, \dots, d\}$. For any non-actionable feature $j \notin \mathcal{A}$, the constraint $\underline{x}_t'^{(j)} = \underline{x}_t^{(j)}$ must hold for all t.

III. Comparative Analysis of Counterfactual Methods

Building upon the formal framework of Section II, we analyze existing counterfactual explanation methods according to the properties defined therein. The analysis focuses on the degree to which each method enforces validity, minimality, temporal plausibility, and actionability, and on the modeling assumptions that condition these guarantees.

A. Model-Dependent Considerations

The properties of counterfactual explanations are deeply influenced by the underlying forecasting model. A first distinction concerns differentiability : gradient-based methods [4], [6] require access to the model's gradient, restricting their applicability to differentiable architectures. Evolutionary [7] and neighborhood-based [8] approaches are model-agnostic and operate on the prediction function alone.

A second consideration concerns the autoregressive structure of forecasting models. Perturbations applied to the input may propagate non-linearly over time, leading to unpredictable effects on multi-step forecasts. This motivates explicit enforcement of temporal plausibility, which naive gradient-based perturbation does not guarantee.

B. ForecastCF

ForecastCF [4] generates counterfactual explanations for univariate time series forecasting by applying gradient descent directly on the input series. Given a black-box differentiable forecasting model f, lower bound α, and upper bound β, the algorithm minimizes the following loss :

$$\mathcal{L} = v \odot \left(\|f(x^*) - \alpha\| + \|\beta - f(x^*)\| \right), \tag{1}$$

where v is a binary masking vector zeroing out timesteps whose forecasted value already satisfies the target bounds. Optimization proceeds via Adam until all forecast timesteps fall within $[\alpha, \beta]$ or the maximum iteration budget is exhausted.

Critically, this loss function contains *no proximity term* : ForecastCF optimizes validity exclusively and does not minimize the perturbation applied to the input. As a consequence, the generated counterfactuals may deviate substantially from the original series, and minimality is not guaranteed in any formal sense. Temporal plausibility is likewise not enforced, as the gradient perturbation modifies each timestep independently without temporal regularization. The method is restricted to differentiable models and deterministic point forecasts.

C. CET-X

CET-X [6] extends the gradient-based approach to multivariate settings by restricting perturbations to exogenous, controllable input variables. Given a univariate target series and a set of exogenous co-variates, CET-X optimizes the following objective :

$$\mathcal{L} = \lambda_1 \|f(x') - y_{\text{target}}\| + \lambda_2 \|e' - e\|, \tag{2}$$

where e and e' denote the original and modified exogenous variables respectively, and λ_1, λ_2 balance validity against minimality. Unlike ForecastCF, CET-X explicitly incorporates a proximity term, providing a formal minimality incentive. Actionability is enforced by construction : only the designated exogenous variables are modified. Temporal plausibility is partially encouraged through regularization on successive exogenous values but is not strictly guaranteed. The method is model-agnostic provided that the forecasting model is differentiable.

D. Genetic Algorithm (Zuin & Veloso)

Zuin and Veloso [7] propose a model-agnostic evolutionary method for multivariate counterfactual generation. The approach first applies Granger causality tests between all variable pairs to identify causally relevant features, reducing the intervention space and preventing implausible perturbations. For each retained variable, a set of quantile regressors is trained to characterize the distribution of plausible future trajectories. A genetic algorithm then searches the space of quantile indices to find a scenario satisfying the target constraint.

The fitness function combines three objectives :

$$o_1 = \|\hat{f}(X') - Y_{\text{goal}}\|, \tag{3}$$

$$o_2 = 1 - \frac{X \cdot X'}{\|X\|\|X'\|}, \tag{4}$$

$$o_3 = -\log P(X' | X_{\text{obs}}), \tag{5}$$

where o_1 minimizes the distance between the counterfactual forecast and the desired outcome, o_2 measures dissimilarity between original and counterfactual trajectories via cosine distance, and o_3 penalizes low-likelihood scenarios. Together, these objectives jointly optimize validity (o_1), an approximate form of minimality (o_2), and temporal plausibility (o_3). The use of quantile regression further constrains the search to distribution-consistent futures, providing strong plausibility guarantees. Actionability is naturally supported, as the genetic algorithm can be restricted to operate on the subset of actionable variables identified by Granger causality.

E. Qualitative Comparison

Table I summarizes the main properties of the four methods.

Table I
COMPARATIVE ANALYSIS OF COUNTERFACTUAL EXPLANATION METHODS. ✓ : EXPLICITLY SUPPORTED; (✓) : PARTIALLY OR INDIRECTLY SUPPORTED; × : NOT ADDRESSED.

Property	**ForecastCF**	**CET-X**	**GA**
Method family	Grad.	Grad.	Evol.
Task	Forec.	Forec.	Forec.
Multivariate inputs	×	(✓)	✓
Multi-step horizon	✓	✓	✓
CF target defined	✓	✓	✓
Validity	✓	(✓)	✓
Minimality	×	✓	(✓)
Temporal plausibility	×	(✓)	(✓)
Actionability	×	✓	(✓)
Model-agnostic	×	(✓)	✓
Probabilistic support	×	×	(✓)
Sparsity	×	(✓)	×

The comparative analysis reveals a fundamental tension in the current literature : methods that guarantee *validity* through explicit optimization, such as ForecastCF and CET-X, often sacrifice *minimality* or *plausibility*. Conversely, approaches that enforce plausibility, like GA [7], introduce significant modeling complexity (e.g., Granger causality tests, quantile regressors). No existing method simultaneously satisfies all desirable properties in a multivariate context.

Among these, ForecastCF stands out as the only gradient-based approach providing a strict, by-construction validity guarantee for univariate point-forecasting. Its model-agnostic nature and compatibility with differentiable architectures make it an ideal candidate for a controlled empirical evaluation. By focusing on ForecastCF, the following sections establish a reproducible baseline for univariate counterfactual forecasting, against which more complex multivariate frameworks can be compared in future work.

IV. Experimental Setup

We evaluate ForecastCF [4] through four sequential experiments (E0-E3), each addressing a targeted research question about its behavior across diverse forecasting settings. All code, pre-trained model checkpoints, and result files are publicly available at https://github.com/kadidiatou131313/forecastcf-univariate-ts-counterfactuals.

A. Research Questions

- **E0.** Which (dataset, model) pairs provide a sufficiently reliable forecasting benchmark to serve as a foundation for counterfactual evaluation?
- **E1.** Does a model's predictive accuracy (R^2) correlate with the quality of the counterfactual explanations it enables?
- **E2.** Does ForecastCF significantly improve counterfactual quality over naive, model-free baseline methods?
- **E3.** How does the width of the target tolerance interval affect counterfactual validity and compactness?

B. Forecasting Pipeline

1) Datasets: To evaluate the robustness of our framework across diverse temporal structures, we selected four datasets characterized by different seasonalities, regularities, and noise levels (see Table II).

Table II
Summary of evaluation datasets.

Dataset	Target Var.	Key Property
ETTh1	HUFL	Strong seasonality
Weather	p	High regularity, multivariate signal
Sunspots	Sunspots	Long-range cycle
Electricity	MT_001	High-dimensional, noisy demand patterns

This selection ensures a broad range of forecasting difficulty, which is essential for Experiment E1 to test whether model performance correlates with counterfactual quality. The datasets cover :

- *ETTh1 :* Industrial series with clear hierarchical cycles.
- *Weather :* Meteorological data providing a regular signal.
- *Sunspots :* Long-range univariate series testing temporal dependencies.
- *Electricity :* Large-scale dataset representing complex demand patterns.

2) Architectures: Three architectures representing complementary modeling families are evaluated, all implemented via the Darts library (see Table III).

By covering linear decomposition, recurrent processing, and pure MLP-based approaches, this selection ensures that results are not specific to one modeling family. All hyperparameters are fixed uniformly across datasets (hidden size 32 for GRU, 2 stacks $\times$ 2 blocks $\times$ 128 units for NBeats), with 30 training epochs and early stopping on the validation set.

Table III
Summary of forecasting models used in the evaluation.

Model	Core Mechanism	Key Property
DLinear	Decomposition into trend and seasonal components via linear layers.	Simple yet competitive baseline; highly interpretable.
GRU	Recurrent processing of temporal steps through gated hidden states.	Captures strong sequential dynamics and autocorrelation.
N-BEATS	Doubly residual stacking of fully connected blocks.	State-of-the-art performance; interpretable by design.

3) Data Split and Evaluation Windows: All series are split temporally in a 60/20/20 ratio (train / validation / test).

Forecasting is performed under a *rolling origin* protocol : a fixed-length input window of size L is slid across the test set to generate successive predictions over a fixed horizon H. Window sizes are dataset-dependent and follow the conventions established in recent long-sequence forecasting benchmarks :

- ETTh1, Weather, Electricity : $L = 96$, $H = 24$ (one hour look-back, one hour ahead for hourly datasets; scaled proportionally for Weather).
- Sunspots : $L = 48$, $H = 12$ (four-year look-back, one-year ahead), chosen to capture at least half a full sunspot cycle in the input window.

The choice of L as a multiple of H ensures that the input window covers at least one full seasonal period for ETTh1 (24-hour daily cycle $\times$ 4 = 96 steps) while remaining computationally tractable for gradient-based CF generation.

4) Forecasting Metrics: Model quality is assessed using three standard regression metrics : mean absolute error (MAE), root mean squared error (RMSE), and the coefficient of determination (R^2). A pair is retained for E1-E3 only if $R^2 \geq 0.50$ on the test set, ensuring that all subsequent counterfactual experiments are grounded in models with genuine predictive signal.

C. Counterfactual Pipeline

1) Counterfactual Metrics: We adopt the four evaluation metrics defined in Wang et al. [4], implemented identically to the original paper to ensure comparability :

- **Validity Ratio** ($\uparrow, \in [0,1]$) : proportion of forecast steps whose counterfactual prediction falls within the target interval :

$$\mathrm{Ratio}(\underline{x}') = \frac{1}{H} \sum_{t=1}^{H} \mathbf{1}[\hat{y}'_t \in [\ell_t,\, u_t]]\,.$$

- **Proximity** ($\downarrow, \geq 0$) : normalized ℓ_2 distance between the original and counterfactual input windows, measuring the magnitude of the perturbation :

$$\mathrm{Proximity}(\underline{x}') = \frac{1}{L} \left\| \underline{x}' - \underline{x} \right\|_2 .$$

- **Compactness** ($\uparrow, \in [0,1]$) : proportion of input timesteps left *unchanged* within a tolerance $\varepsilon > 0$, measuring the sparsity of the perturbation :

$$\text{Compact}(\underline{x}') = \frac{1}{L}\sum_{t=1}^{L} \mathbf{1}[|x'_t - x_t| \leq \varepsilon]\,.$$

- **Step-AUC** ($\uparrow, \in [0,1]$) : area under the curve of the *consecutive* validity profile over the forecast horizon. For each step $h \in \{1,\ldots,H\}$, the stepwise validity is the proportion of the first h forecast steps that fall within the target interval; Step-AUC is the normalized integral of this curve. This metric rewards contiguous validity and penalizes methods that satisfy the target only on isolated steps [4].

2) Counterfactual Window Selection: For each (dataset, model) pair, $N = 10$ input windows are sampled uniformly at random from the test set without replacement, and CF metrics are averaged across all 10 instances. This sampling strategy avoids temporal position bias : sequential sampling from the beginning of the test set would over-represent a specific temporal regime (warm-up, seasonal peak, etc.) and could bias CF metrics toward windows that are structurally easier or harder to perturb.

3) Target Bound Formulation: Following Wang et al. [4], target bounds are defined adaptively :

$$\ell = \hat{y} + \delta \cdot \sigma_x, \qquad u = \hat{y} + (\delta + w) \cdot \sigma_x,$$

where $\hat{y}$ is the original forecast, σ_x is the standard deviation of the input window, $\delta = 0.5$ is a fixed shift, and $w = 0.5$ is the tolerance width (varied in E3). Normalizing by σ_x ensures comparability across datasets with different scales.

4) Baselines (E2): ForecastCF is compared against two naive baselines introduced in the original paper [4] to quantify the marginal contribution of gradient-based optimization :

ShiftCF applies a fixed multiplicative shift to the entire input window, requiring no model access. It is the simplest possible perturbation strategy and runs in near-zero time.

NNCF retrieves the nearest neighbor from the training set whose model prediction lies closest (in ℓ_2 distance) to the center of the target interval. It requires an offline indexing step over training predictions but no gradient computation at inference time.

V. Experimental Results

A. E0 - Reference Benchmark

Table IV
E0 - Forecasting benchmark results on the test set. All 12 pairs exceed the $R^2 \geq 0.50$ retention threshold (✓).

Dataset	Model	MAE	RMSE	R^2	
ETTh1	DLinear	0.544	0.810	0.692	✓
	GRU	0.312	0.453	0.904	✓
	NBeats	0.553	0.802	0.698	✓
Weather	DLinear	0.054	0.079	0.996	✓
	GRU	0.007	0.010	1.000	✓
	NBeats	0.056	0.081	0.995	✓
Sunspots	DLinear	0.493	0.680	0.784	✓
	GRU	0.341	0.472	0.894	✓
	NBeats	0.500	0.682	0.783	✓
Electricity	DLinear	0.382	0.734	0.636	✓
	GRU	0.215	0.431	0.874	✓
	NBeats	0.350	0.693	0.675	✓

All 12 (dataset, architecture) pairs exceed the $R^2 = 0.50$ threshold, with values spanning 0.636 to 0.9999 (see in table IV).

GRU achieves the best R^2 on three of four datasets (ETTh1 : 0.904, Electricity : 0.874, Sunspots : 0.894), confirming the advantage of recurrent architectures on series with strong temporal dependencies.

Weather is a degenerate case : all three models exceed $R^2 > 0.995$, reflecting the unusually high regularity of atmospheric pressure.

The wide spread of R^2 across pairs (0.636 to 0.9999) provides the distributional range required to test the correlation hypothesis in E1.

A bar chart of all 12 R^2 values sorted by performance, colored by architecture, is provided in the appendix, together with individual forecast visualizations for each pair.

B. E1 - Forecasting Performance vs. Counterfactual Quality

Table V
E1 - CF METRICS AVERAGED OVER 10 RANDOM TEST INSTANCES PER PAIR (MEAN ± STD). SPEARMAN CORRELATIONS WITH R^2 : VALIDITY ρ=−0.078 (p=0.81), PROXIMITY ρ=−0.399 (p=0.20), COMPACTNESS ρ=0.000 (p=1.00).

Dataset	Model	Validity	Prox. L2	Compact.
3*ETTh1	DLinear	1.000±.000	8.28±1.90	.031±.019
	GRU	0.525±.501	8.60±3.19	.007±.014
	NBeats	0.963±.119	6.55±2.05	.060±.155
3*Electricity	DLinear	1.000±.000	2.37±1.52	.005±.006
	GRU	0.348±.154	3.57±1.97	.076±.054
	NBeats	0.983±.053	3.18±1.64	.121±.226
3*Sunspots	DLinear	1.000±.000	2.16±1.30	.004±.009
	GRU	1.000±.000	5.43±2.63	.000±.000
	NBeats	0.958±.132	3.94±2.42	.050±.094
3*Weather	DLinear	1.000±.000	0.23±0.13	.541±.121
	GRU	1.000±.000	1.15±0.50	.000±.000
	NBeats	0.921±.107	1.01±1.20	.446±.300

Spearman rank correlation between R^2 and CF metrics yields no significant association across any metric (Table V, caption). This negative result is the central finding of E1 : **forecasting accuracy is not a predictor of counterfactual quality**.

The (ETTh1 , GRU) pair illustrates the underlying mechanism. Despite achieving the highest R^2 (0.904), it produces the lowest Validity (0.525). A highly accurate GRU develops sharp non-linear dynamics that destabilize ForecastCF's gradient descent, requiring large perturbations to reach the target.

Compactness is near zero for most pairs : ForecastCF systematically modifies nearly all input timesteps, confirming that sparsity is the hardest property for gradient-based perturbation to achieve. This is consistent with the original results reported by Wang et al. [4].

Scatter plots of R^2 vs. each CF metric with pair labels are provided in figure 2, and the per-pair metric bar charts are provided in the appendix.

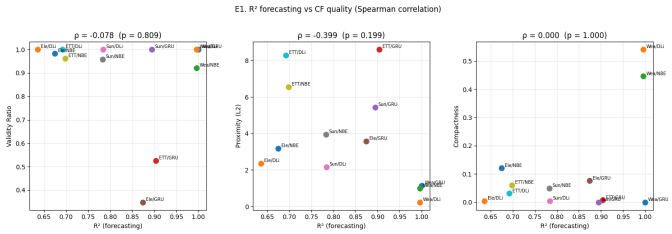

Figure 2. E1 - Spearman correlation between R^2 (test) and counterfactual quality metrics across all 12 pairs. No significant association is found for any metric (all $p > 0.19$).

C. E2 - Comparison with Naive Baselines

Table VI
E2 - WILCOXON SIGNED-RANK TESTS (n=120 POOLED INSTANCES). FORECASTCF VS. EACH BASELINE ON ALL METRICS. * $p < 0.05$.

Metric	Comparison	p-value	
2*Validity ↑	vs. ShiftCF	< 0.0001	*
	vs. NNCF	< 0.0001	*
2*Proximity ↓	vs. ShiftCF	0.0085	*
	vs. NNCF	< 0.0001	*
2*Compactness ↑	vs. ShiftCF	< 0.0001	*
	vs. NNCF	0.0056	*
2*Time (s) ↓	vs. ShiftCF	< 0.0001	*
	vs. NNCF	< 0.0001	*

ForecastCF significantly outperforms both baselines on all four metrics (Wilcoxon signed-rank, $p < 0.05$ in every case, see in table VI).

On **Validity**, the advantage over ShiftCF and NNCF is categorical ($p < 0.0001$ both) : gradient descent explicitly guides perturbation toward the target interval, whereas ShiftCF shifts the input blindly and NNCF retrieves a training example with no local optimization.

On **Proximity**, ForecastCF is significantly closer to the original than NNCF ($p < 0.0001$); the smaller advantage over ShiftCF ($p = 0.0085$) reflects the fact that a uniform multiplicative shift is itself a compact operation, but one that sacrifices validity.

On **Compactness**, the Wilcoxon test detects a significant difference between ForecastCF and both baselines ($p < 0.0001$ vs. ShiftCF; $p = 0.0056$ vs. NNCF), yet all three methods achieve near-zero compactness. This reflects the structural limitation of gradient-based perturbation identified in E1 : without an explicit sparsity term in the loss, ForecastCF modifies nearly every input timestep, and the two baselines behave similarly for different reasons - ShiftCF shifts all points uniformly by construction, and NNCF returns a training instance that differs from the query across the full window.

The **generation time** of ForecastCF is significantly higher than both baselines ($p < 0.0001$), ranging from < 0.01s (DLinear/Electricity) to ~ 8s per instance (GRU/Electricity). This cost is the main operational drawback of the method relative to model-free alternatives.

Fig. 3 provides a Critical Difference (CD) diagram for each metric, showing average ranks across all 12 (dataset, model) pairs. Methods connected by a bold bar are not significantly different. ForecastCF ranks first on Validity (avg. rank 1.00) and Proximity (1.25).

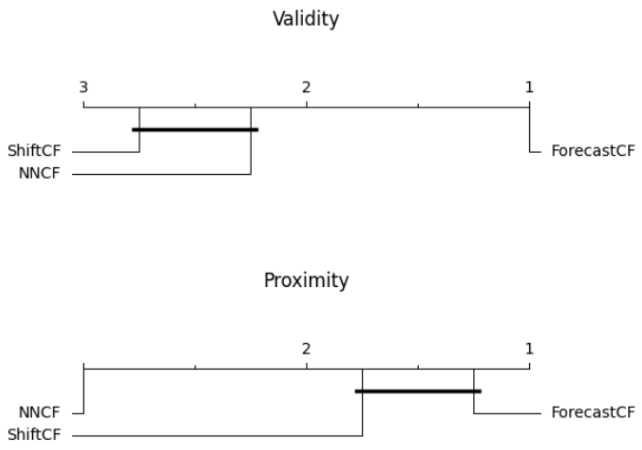

Figure 3. E2 - Critical Difference (CD) diagrams for Validity and Proximity across 12 pairs. Methods connected by a bold bar are not significantly different. ForecastCF ranks first on Validity and Proximity.

Grouped bar charts of all metrics per (dataset, model) pair are provided in Fig. 3 and in the appendix.

D. E3 - Sensitivity to Target Tolerance Width

Increasing the tolerance width w from 0.1σ to 0.6σ produces consistent effects across all 12 pairs.

Validity increases with width for almost all pairs, as expected : a wider interval is geometrically easier to reach by gradient descent. Some pairs remain problematic : ETTh1/GRU starts from Validity ≈ 0.01 at 0.1σ and reaches only 0.62 at 0.6σ; Electricity/GRU similarly plateaus below 0.40. These pairs reveal a structural difficulty of recurrent architectures for tight constraints, consistent with the high-sensitivity finding of E1.

Proximity increases with width in most cases : a wider band placed further from the original prediction requires a larger shift of the input to reach the center of the target. Non-monotonic exceptions appear for Electricity/GRU and Sunspots/NBeats at 0.3σ, likely due to the small sample size ($N = 5$) in E3.

Compactness decreases with width, confirming the fundamental validity-compactness trade-off : achieving a wider (easier) target requires modifying more input timesteps.

A width of $w = 0.3\sigma$ provides the best practical compromise : Validity exceeds 0.60 for 10 of 12 pairs without causing severe compactness degradation, aligning with Wang et al.'s recommendation to avoid bounds that are too tight ($w < 0.25\sigma$) or too loose ($w > 1\sigma$) [4].

Fig. 4 shows line plots of Validity and Compactness vs. w for two representative pairs. The full 12-pair grid for all three metrics is provided in the appendix.

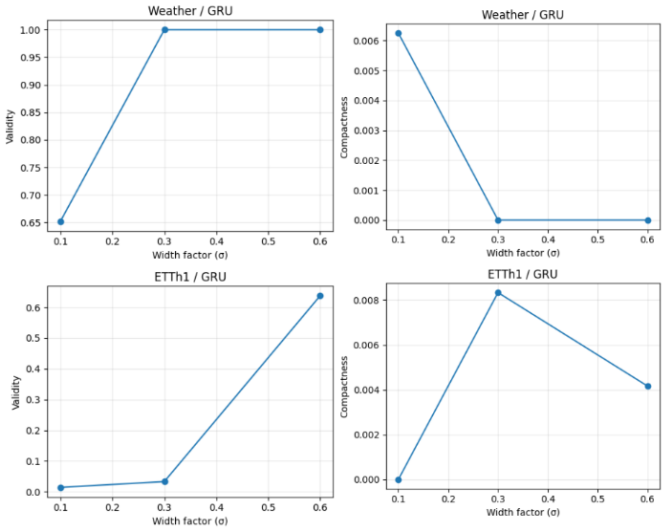

Figure 4. E3 - Validity (left) and Compactness (right) as a function of tolerance width w. Wider tolerances consistently increase Validity and decrease Compactness across architectures.

VI. Discussion

A. Comparison with Wang et al.'s Original Results

Our results closely replicate and extend the empirical findings of Wang et al. [4]. In the original paper, ForecastCF achieves near-perfect Validity Ratio on several datasets and consistently outperforms the NNCF baseline on proximity. Our evaluation confirms these findings on a broader set of 12 (dataset, model) pairs and introduces two additional contributions : (i) the Wilcoxon significance tests, which formalize the observed differences between methods, and (ii) the correlation analysis in E1, which explicitly tests whether R^2 predicts CF quality - a question not addressed in the original work.

The near-zero compactness scores observed across all our pairs are also consistent with Wang et al.'s discussion : they report that ForecastCF does not include a sparsity regularization term in its loss function, which mechanistically explains the dense perturbation pattern we observe. As formally established in Section III, the loss $\mathcal{L}$ contains no proximity or sparsity term, optimizing validity exclusively. This design choice is a key limitation that our unified framework can now clearly articulate through the compactness property.

B. Key Findings and Practical Implications

The accuracy-explainability gap (E1). The absence of correlation between R^2 and CF quality has a direct practical consequence : practitioners cannot rely on standard model selection criteria to ensure that the chosen model will support high-quality explanations. This implies that CF quality should be evaluated as an *independent* axis of model assessment, orthogonal to predictive performance. Our formal framework provides the vocabulary to do so consistently.

The necessity of gradient guidance (E2). ShiftCF, which is orders of magnitude faster, achieves near-zero Validity across most pairs. NNCF achieves moderate Validity at the cost of

high Proximity. Neither baseline offers the validity-proximity balance that ForecastCF provides.

Tolerance as a practitioner hyperparameter (E3). The recommendation of $w \approx 0.3\sigma$ is robust across datasets and linear/MLP architectures, but not for recurrent models under tight constraints. In practice, this suggests adapting w to the architecture sensitivity observed in E0, rather than fixing it globally.

C. Limitations

The evaluation is restricted to the univariate point-forecasting setting. ForecastCF perturbs all timesteps of a single target series without regard for actionability or cross-variable structure. The formal framework of Section II exposes this gap explicitly : actionability and temporal plausibility properties are never enforced by ForecastCF.

The compactness problem is structural : without an explicit sparsity term in the loss, gradient descent modifies the entire input window. Addressing this would require a regularization term that penalizes $\|X' - X\|_0$, which is non-differentiable.

D. Towards Multivariate Evaluation with GA

The GA approach of Zuin and Veloso [7] is the natural next candidate for comparison, for three reasons that map directly onto the limitations above.

First, GA is *model-agnostic* : it does not require differentiable models, making it applicable to any black-box forecaster, including tree-based methods and hybrid architectures outside the Darts ecosystem.

Second, GA explicitly addresses temporal plausibility through likelihood-based fitness objectives and Granger causality constraints, which ForecastCF leaves entirely to the gradient dynamics. A direct comparison would quantify the plausibility gap using the temporal increment criterion, which was formalized in our framework but not yet empirically evaluated.

Third, GA supports actionability by restricting interventions to causally relevant variables. On the multivariate datasets already available in our benchmark (ETTh1 : 7 variables, Weather : 21 variables), this would correspond to modifying only co-variates with established causal influence on the target, a critical property for high-stakes applications.

The expected outcome of such a comparison is a validity-plausibility trade-off : ForecastCF should achieve higher Validity thanks to strict gradient convergence, while GA should produce temporally smoother and more actionable counterfactuals. Testing this hypothesis on causally structured data (Electricity, where demand drivers are partially observable) would constitute a systematic evaluation of temporal plausibility in counterfactual generation for forecasting.

VII. Conclusion

This paper made three contributions to the study of counterfactual explanations for time series forecasting.

We first proposed a unified formal framework covering validity, minimality, temporal plausibility, and actionability for continuous-valued prediction settings. This framework provides a principled basis for comparing methods that have so far been evaluated under incompatible assumptions.

We then conducted a structured qualitative comparison of representative methods (ForecastCF [4], CET-X [6], GA[7]), revealing a fundamental tension between methods that guarantee validity through explicit optimization and methods that enforce temporal plausibility through distributional constraints.

Finally, we carried out a systematic empirical evaluation of ForecastCF across 4 datasets and 3 architectures.

Future work will extend the evaluation to the multivariate setting through comparison with the GA approach, incorporate temporal plausibility as an explicit optimization constraint in ForecastCF, and explore probabilistic counterfactual targets for interval and quantile forecasting models.

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